

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Principals Of Electrical Engg. Course Code: EE-101

Examination: B.Tech Scheme: CBS Branch: CSE,ECE,CE Semester: 1st

Question Paper for 2nd Periodical Examinations (Nov, 2023)

Time: Two Hours

Max. Marks: 50

Course Outcomes:-

CO101.1. Recalling Basic Electrical and magnetic laws

CO101.2 Remembering Steady State analysis of DC circuits

CO104.3 Understanding AC Single phase & Three phase Circuits

CO104.4 Solving complex AC, DC circuits

CO104.5 Explaining various concepts of Electrical machines

Q1. (CO101.3) Explain in detail the measurement of power by 2 wattmeter method? **(10)**

Q2. (CO101.5) Explain core and shell type of transformer with the help of diagram? **(10)**

Q3. (CO101.2) Derive the expression for Transient analysis of RC series circuit with DC excitation (charging & discharging) with the help of diagram? **(10)**

Q4. (CO101.3) Explain star connected 3-phase system in detail? Derive the relation between phase voltage & line voltage in case of star system? **(10)**

Q2 (CO101.4)

a) Two wattmeter method is used to measure the power taken by 3 phase induction motor on no load. The wattmeter readings are 375 W & -50 W. Calculate

I. Power factor at no load

II. Phase difference of voltage & current in two wattmeters **(5)**

b) Calculate time taken by a capacitor of 1 microfarad & in series with a 1 mega ohm resistance to be charged upto 80% of final value? **(5)**

(Course Advisor: Manavi Sharma)